Practical No: 1

❖ **Aim: What Is MySQL And Database?**

- MySQL:
  a) MySQL is a relational database management system.
  b) MySQL is open-source.
  c) MySQL is free.
  d) MySQL is ideal for both small and large applications.
  e) MySQL is very fast, reliable, scalable, and easy to use.
  f) MySQL is cross-platform.
  g) MySQL is compliant with the ANSI SQL standard.
  h) MySQL was first released in 1995.
  i) MySQL is developed, distributed, and supported by Oracle Corporation.
  j) MySQL is named after co-founder Ulf Michael "Monty" Widenius's daughter: My.
- Database system:
  a) DBMS allows users to define the structure of the database, including the types of data it will hold and how it will be organized (e.g., tables, rows, columns).
  b) **Data Security:** DBMS implements security measures to control access to the database and protect it from unauthorized use.
  c) **Data Integrity:** DBMS ensures data consistency and accuracy through various mechanisms, like constraints and validation rules.
  d) **Concurrency Control:** DBMS manages multiple users accessing and modifying data simultaneously, preventing conflicts and maintaining data integrity.
  e) **Data Backup and Recovery:** DBMS provides tools for backing up the database and restoring it in case of failure.
  f) Why Use a DBMS?
  g) **Efficient Data Managed:** DBMS provides a structured and efficient way to store, organize, and retrieve data, especially when dealing with large datasets.
  h) **Reduced Data Redundancy:** DBMS helps minimize data duplication, which can lead to inconsistencies and storage inefficiencies.

i) **Improved Data Integrity:** DBMS enforces rules and constraints to ensure data accuracy and consistency.
j) **Enhanced Security:** DBMS provides mechanisms to control access to data and protect it from unauthorized access.
k) **Simplified Data Access:** DBMS provides tools and interfaces for users to easily access and manipulate data.
l) **Support for Concurrent Access:** DBMS allows multiple users to access and modify data simultaneously without conflicts.
m) Examples of DBMS:
   I. **Relational DBMS (RDBMS):** MySQL, PostgreSQL, Oracle, Microsoft SQL Server.
   II. **NoSQL DBMS:** MongoDB, Cassandra.

- RDBMS:

a) RDBMS Stands For Relational Database Management System.

b) RDBMS is a program used to maintain a relational database.

c) RDBMS is the basis for all modern database systems such as MySQL, Microsoft SQL Server, Oracle, and Microsoft Access.

d) RDBMS uses SQL queries to access the data in the database.

e) A relational database management system (RDBMS) is a type of database manage that stored data in structured format using raw and column.

- Advantage of RDBMS:

a) Data integrity and accuracy.

b) Easy to use and query using SQL.

c) Reduces redundancy.

d) Support relationship between data (table).

e) Scalable and secure.

- Schema:

  a) A database schema is a description of the logical structure of a database or the element it contains.

  b) Schema often include dedications for the structure of individual attribute that database entries are comprised.

  c) These may also define data type and additional constraints to control the type of data that may be added to the structure.

SQL COMMANDS			
DDL	**DML**	**DCL**	**TCL**
CREATE	SELECT	GRANT	COMMIT
ALTER	DELETE	REVOKE	ROLLBACK
DROP	INSERT		SAVEPOINT
RENAME	UPDATE		SET-TRANSACTION
COMMENT	CALL		
TRUNCATE	MARGE		
	EXPLAIN PLAN		
	LOCK TABLE		

Practical No: 2

- **Aim: to create database:-**
- **Aim: To Create a Table with Various Columns of Different Data-types to record Sales of a Product**.

  - create database tanmay2025_26;
  - use tanmay2025_26;
  - create table unit_sales(SID int primary key,pname varchar(30),
  - Qty_units int,DOS date, S_no int);
  - desc unit_sales;

- **Output:**

Field	Type	Null	Key	Default	Extra
SID	int	NO	PRI	NULL	
pname	varchar(30)	YES		NULL	
Qty_units	int	YES		NULL	
DOS	date	YES		NULL	
S_no	int	YES		NULL	

  - insert into unit_sales values(1,'laptop',5,'2025-04-24',101),
  - (2,'keyboard',15,'2025-11-02',102),
  - (3,'mouse',20,'2025-01-31',103);
  - select*from unit_sales;

- **Output:**

SID	pname	Qty_units	DOS	S_no
1	laptop	5	2025-04-24	101
2	keyboard	15	2025-11-02	102
3	mouse	20	2025-01-31	103
NULL	NULL	NULL	NULL	NULL

Practical No: 3

❖ **Aim: To Create A Table To Record Data Of Employee.**

- create database tanmay2025_26;
- use tanmay2025_26;
- create table employee(Eid int primary key, name varchar(30),
- DOB date,Gender varchar(1),
- Dept varchar(30));
- desc employee;

- **Output:**

Field	Type	Null	Key	Default	Extra
Eid	int	NO	PRI	NULL	
name	varchar(30)	YES		NULL	
DOB	date	YES		NULL	
Gender	varchar(1)	YES		NULL	
Dept	varchar(30)	YES		NULL	

❖ **Aim: inserting to values in the employee table and using various commands of alter in it.**

- insert into employee
- (Eid,name,DOB,Gender,Dept)values
- (101,'Rahul','2003-11-17','M','B.SC(CS)'),
- (102,'Pawan','2005-09-30','M','B.SC(IT)'),
- (103,'Tanmay','2005-10-20','M','B.A'),
- (104,'Riya','2007-08-23','F','B.COM');
- select*from employee;

- **Output:**

Eid	name	DOB	Gender	Dept
101	Rahul	2003-11-17	M	B.SC(CS)
102	Pawan	2005-09-30	M	B.SC(IT)
103	Tanmay	2005-10-20	M	B.A
104	Riya	2007-08-23	F	B.COM
NULL	NULL	NULL	NULL	NULL

- alter table employee add(sal decimal(7,2));
- alter table employee add sal decimal(7,2) after Gender;
- alter table employee modify Dept varchar(10);

- desc employee;
- alter table employee modify sal decimal(8,2)default 20000;
- desc employee;
- alter table employee change DOB birth_date date;
- desc employee;
- delete from employee where Eid=104;
- select*from employee;
- update employee set Dept='CS' where Eid=102;
- select*from employee;

- **Output:**

Eid	name	birth_date	Gender	Dept	sal
101	Rahul	2003-11-17	M	B.SC(CS)	NULL
102	Pawan	2005-09-30	M	CS	NULL
103	Tanmay	2005-10-20	M	B.A	NULL
NULL	NULL	NULL	NULL	NULL	NULL

Practical No: 4

❖ **Aim: To Create A Simple Table For Recording Data Of Employees With Various Data Attributes.**

- create table Employees
- (Eno int primary key, Name varchar(30),
- DOB date, Gender varchar(1),dept varchar(5),
- Salary decimal(7,2));
- desc employees;

❖ **Aim: To Insert Values In The Table.**

- insert into Employees(Eno,Name,DOB,Gender,dept,salary)
- values(001,'Jack','1999-05-31','M','MBA','35000.00'),
- (002,'Nora','2000-07-12','F','CA','40000.00');
- select*from Employees;

- **Output:**

Eno	Name	DOB	Gender	dept	Salary
1	Jack	1999-05-31	M	MBA	35000.00
2	Nora	2000-07-12	F	CA	40000.00
NULL	NULL	NULL	NULL	NULL	NULL

❖ **Aim: Altering The Pre-Existing Table.**

- alter table Employees add father_name varchar(20) after Gender;
- alter table Empolyees modify dept varchar(10);
- alter table Employees modify salary decimal(8,1) default 40000;
- alter table Employees rename column Gender to Sex;
- select*from Employees;

❖ **Aim: To View The Table:**

- **Output:**

Eno	Name	DOB	Sex	father_name	dept	Salary
1	Jack	1999-05-31	M	NULL	MBA	35000.00
2	Nora	2000-07-12	F	NULL	CA	40000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Practical No: 5

❖ **Aim: To Create And Using Database.**

- create database tanmay2025_2026;
- use tanmay2025_2026;

❖ **Aim: To Create A Table To Record Data On Branches With Columns With Various Data Attributes.**

- create table branch2 (br_code smallint unsigned not null, location varchar(20),
- zone varchar(10), br_head int, target decimal(8,2),actual decimal(8,2),
- constraint pk_br_code primary key (br_code));
- desc branch2;

❖ **Aim: To Insert Data In The Column.**

- insert into branch2
- values(15,'andheri','western',105,120000,150000),
- (20,'badlapur','eastern',103,220000,170000),
- (25,'dombivali','western',104,100000,150000),
- (30,'kalyan','eastern',106,190000,100000);
- select*from branch2;

- **Output:**

br_code	location	zone	br_head	target	actual
15	andheri	western	105	120000.00	150000.00
20	badlapur	eastern	103	220000.00	170000.00
25	dombivali	western	104	100000.00	150000.00
30	kalyan	eastern	106	190000.00	100000.00
NULL	NULL	NULL	NULL	NULL	NULL

- create table staff(st_no smallint unsigned,name varchar(20),age smallint unsigned,
- rep_br smallint unsigned, desg varchar(10), join_date date, boss_id int, sex char(1),

- constraint pk_st_no primary key (st_no),
- constraint fk_br_code foreign key (rep_br) references
- branch2(br_code));
- desc staff;
- insert into staff values
- (105,'tania gomes',24,15,'officer','2002-12-10',105,'f'),
- (109,'nora hasnani',23,20,'actn','2005-06-30',104,'f'),
- (102,'muskan gupta',21,25,'actn','2006-07-14',106,'f'),
- (106,'sancia dsa',30,20,'manager','1997-12-09',103,'f'),
- (104,'shoydon alphanso',20,30,'officer','2007-09-23',104,'m'),
- (101,'devang vanmali',23,25,'actn','2004-02-14',104,'m'),
- (110,'tanmay hindlekar',21,15,'agent','2009-01-08',101,'m'),
- (108,'rahul mahdik',26,25,'officer','2004-03-21',106,'m'),
- (103,'raj kumar',24,30,'actn','2008-11-05',104,'m'),
- (107,'jenny roy',25,15,'actn','2006-02-11',108,'f');
- select*from staff;

- **Output:**

st_no	name	age	rep_br	desg	join_date	boss_id	sex
101	devang vanmali	23	25	actn	2004-02-14	104	m
102	muskan gupta	21	25	actn	2006-07-14	106	f
103	raj kumar	24	30	actn	2008-11-05	104	m
104	shoydon alphanso	20	30	officer	2007-09-23	104	m
105	tania gomes	24	15	officer	2002-12-10	105	f
106	sancia dsa	30	20	manager	1997-12-09	103	f
107	jenny roy	25	15	actn	2006-02-11	108	f
108	rahul mahdik	26	25	officer	2004-03-21	106	m
109	nora hasnani	23	20	actn	2005-06-30	104	f
110	tanmay hindlekar	21	15	agent	2009-01-08	101	m
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Practical No: 6

- **Aim: To Create A Table For Recording Data Of Branches With Primary Key Constraint.**
  - create table Branches1
  - (Br_code int not null, Location varchar(20),
  - zone varchar(10),
  - BR_HEAD int, target decimal(10,2),
  - actual decimal(10,2),
  - constraint pk_Br_code primary key(Br_code));
  - desc Branches1;
- **Aim: To Create Two Tables For Branches And Employees And Declaring Foreign Key In The Second Table.**
  - insert into Branches1
  - (Br_code,Location,zone,BR_HEAD,target,actual)
  - values(22,'borivali','western', 106,200000,170000),
  - (34,'mulund','central', 104,260000,280000),
  - (20,'parle','western', 105,330000,330000),
  - (25,'thane','central', 108,250000,170000),
  - (18,'vasai','western', 108,25000,null),
  - (21,'malad','western', 106,400000,400000);
  - select*from Branches1;
- **Output:**

Br_code	Location	zone	BR_HEAD	target	actual
18	vasai	western	108	25000.00	NULL
20	parle	western	105	330000.00	330000.00
21	malad	western	106	400000.00	400000.00
22	borivali	western	106	200000.00	170000.00
25	thane	central	108	250000.00	170000.00
34	mulund	central	104	260000.00	280000.00
NULL	NULL	NULL	NULL	NULL	NULL

Practical No: 7

❖ **Aim: To Create Two Different Tables And Combining Them By Using Inner join.**

- use tanu12;
- create table students(id integer primary key not null,
- name varchar(20), age integer, address varchar(30));
- desc students;
- insert into students(id,name,age,address)
- values (101,"Tanmay",19,"jogeshwari"),
- (102,"Rahul",18,"andheri"),
- (103,"Raj",19,"dadar"),
- (104,"Yesh",17,"ghatkopar");
- select*from students;
- use tanu12;
- create table admission (id integer not null,
- students_id integer, course_name varchar(10),
- constraint pk_id primary key(id),
- constraint fk_students_id foreign key (students_id)
- references students(id));
- insert into admission
- (id, students_id, course_name)
- values (1,101,"data sci"),
- (2,101,"fea eng"),
- (3,102,"big data"),
- (4,102,"prompt eng");
- select*from admission;
- select students.name,admission.course_name from students
- inner join admission on students.id=admission.students_id;

- **Output:**

name	course_name
Tanmay	data sci
Tanmay	fea eng
Rahul	big data
Rahul	prompt eng

Practical No: 8

- **Aim: To Create A Two Tables To Record Branches And Employees With Various Data Attribute.**

```
create database hindlekar2025;
use hindlekar2025;
create table branches11(br_code smallint unsigned not null,
location varchar(20), zone varchar(10), br_head int (3),
target decimal(10,2),actual decimal(10,2),
constraint pk_br_code primary key(br_code));
desc branches11;
insert into branches11
values(22,'jogeshwari','eastern',103,200000,170000),
(34,'andheri','eastern',102,250000,170000),
(20,'mulund','wastern',104,330000,300000),
(25,'dadar','eastern',105,200000,222000),
(18,'thane','wastern',107,130000,10000);
select*from branches11;
use hindlekar2025;
create table empl
(e_no smallint unsigned, name varchar(20),
age smallint unsigned, rep_br smallint unsigned, desg varchar(10),
join_date date, boss_id int (6), gender char(1),
constraint pk_e_no primary key(e_no),
constraint fk_br_code foreign key(rep_br)
references branches11(br_code));
desc empl;
insert into empl
values(101,'tanmay hindlekar',19,22,'manager','2015-04-23',103,'m'),
(102,'raj kumar',20,25,'officer','2017-11-23',102,'m'),
(103,'yesh yadav',34,null,'officer','2020-04-22',105,'m'),
(104,'jitesh ram',22,20,'manager','2009-12-17',104,'m'),
```

- (105,'salman khan',19,18,'actn','2016-06-02',null,'m'),
- (106,'vijay pawar',25,34,'actn','2014-08-31',101,'m');
- select*from empl;

- **Output:**

e_no	name	age	rep_br	desg	join_date	boss_id	gender
101	tanmay hindlekar	19	22	manager	2015-04-23	103	m
102	raj kumar	20	25	officer	2017-11-23	102	m
103	yesh yadav	34	NULL	officer	2020-04-22	105	m
104	jitesh ram	22	20	manager	2009-12-17	104	m
105	salman khan	19	18	actn	2016-06-02	NULL	m
106	vijay pawar	25	34	actn	2014-08-31	101	m
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

- select count(*),desg how_many from empl group by desg;

- **Output:**

count(*)	how_many
2	manager
2	officer
2	actn

- select age,count(*) from empl group by age;

- **Output:**

age	count(*)
19	2
20	1
34	1
22	1
25	1

- select gender,count(*) from empl group by gender;

- **Output:**

gender	count(*)
m	6

Practical No: 9

❖ **Aim: To Create A Transaction Table.**

- create database food2025;
- use food2025;
- create table party
- (GST_ID int, Name varchar(20),
- Gend enum('M','F') not null,
- food_pref enum('veg','non-veg') not null,
- primary key(GST_ID));
- desc party;
- insert into party(GST_ID,Name,Gend,food_pref)
- values(1,'rahul kumar','M','veg'),
- (2,'bheem raj','M','veg'),
- (3,'salman khan','M','non-veg'),
- (4,'daya gada','F','veg'),
- (5,'jack saprrow','M','non-veg'),
- (6,'alia bhat','F','non-veg');
- select*from party;

- **Output:**

GST_ID	Name	Gend	food_pref
1	rahul kumar	M	veg
2	bheem raj	M	veg
3	salman khan	M	non-veg
4	daya gada	F	veg
5	jack saprrow	M	non-veg
6	alia bhat	F	non-veg
NULL	NULL	NULL	NULL

- start transaction;
- update party set food_pref='non-veg'where GST_ID=1;
- start transaction;
- update party set food_pref='veg'where GST_ID=6;
- select*from party;

- rollback;
- select*from party;
- start transaction;
- delete from party where GST_ID=4;
- select*from party;
- rollback;
- select*from party;
- start transaction;
- set sql_safe_updates=0;
- delete from party;
- set sql_safe_updates=1;
- select*from party;
- rollback;

- **Output:**

GST_ID	Name	Gend	food_pref
1	rahul kumar	M	non-veg
2	bheem raj	M	veg
3	salman khan	M	non-veg
4	daya gada	F	veg
5	jack saprrow	M	non-veg
6	alia bhat	F	veg
NULL	NULL	NULL	NULL

Practical No: 10

❖ **Aim: Inner Join Queries.**

- create database tanmayfycs;
- use tanmayfycs;
- create table office(off_code smallint unsigned not null,
- location varchar(20),zone varchar(10),off_head int, target decimal(10,2),
- actual decimal(10,2), constraint pk_off_code primary key(off_code));
- desc office;
- insert into office values(19,'jogeshwari','eastern',102,200000,160000),
- (20,'andheri','eastern',101,300000,260000),
- (22,'vasai','wastern',104,220000,190000),
- (16,'thane','central',103,100000,90000),
- (25,'dadar','waetern',105,250000,180000);
- select*from office;
- use tanmayfycs;
- create table staff(st_no smallint unsigned, name varchar(10),
- age smallint unsigned, rep_off smallint unsigned, desg varchar(10),
- join_date date, boss_id int(5),sex char(1),
- constraint pk_st_no primary key (st_no),
- constraint fk_off_code foreign key(rep_off)references
- office(off_code));
- desc staff;
- insert into staff
- values(101,'tanmay',20,19,'managar','2020-02-05',102,'m'),
- (102,'tonny',22,20,'agent','2018-02-15',105,'m'),
- (103,'jack',21,22,'agent','2022-10-10',104,'m'),
- (104,'nova',24,16,'managar','2023-02-11',103,'f'),
- (105,'johnson',28,25,'actn','2016-02-02',101,'m'),
- (106,'maggi',30,19,'officer','2020-10-25',106,'f');
- select*from staff;

- select name,location from staff, office where rep_off=off_code;
- **Output:**

	name	location
▶	nova	thane
	tanmay	jogeshwari
	maggi	jogeshwari
	tonny	andheri
	jack	vasai
	johnson	dadar

- select location,target,name off_head from office inner join staff on off_head=st_no;
- **Output:**

	location	target	off_head
▶	thane	100000.00	jack
	jogeshwari	200000.00	tonny
	andheri	300000.00	tanmay
	vasai	220000.00	nova
	dadar	250000.00	johnson

- select name, desg,location,join_date from staff inner join office on rep_off=off_code;
- **Output:**

	name	desg	location	join_date
▶	nova	managar	thane	2023-02-11
	tanmay	managar	jogeshwari	2020-02-05
	maggi	officer	jogeshwari	2020-10-25
	tonny	agent	andheri	2018-02-15
	jack	agent	vasai	2022-10-10
	johnson	actn	dadar	2016-02-02